Description of Targets in MAP Growth Calculator

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Introduction

The MAP Growth Calculator is a tool that allows users to set a beginning grade and MAP RIT score, and then calculate the student progress that would result from three different rates of growth over the following three test cycles. The student growth can be plotted against one of four different targets. Depending on the rate of growth and the specific target selected, users can see whether a student at a given starting RIT value is meeting or eventually will meet the chosen MAP target.

Targets

The application currently contains four different targets:

1. NWEA Median. This value shows the RIT score at the 50th percentile for fall and spring by grade level based on the most recent NWEA norming study (Northwest Evaluation Association, 2012).
2. MCA Proficiency. This target identifies the MAP RIT scores that predict a strong likelihood of meeting the proficiency score on the spring Minnesota Comprehensive Assessments (MCA) by grade. These targets are based on an analysis of several thousand Minnesota students who took both the MAP and the MCA is the same grade and year. Separate empirical studies have been completed for the MCA-II in Reading (Silberglitt, 2008) and the MCA-III in Math (Silberglitt & Muyskens, 2012). [i]
3. College/Career Readiness. College readiness benchmark scores have previously been established by ACT® from empirical data to identify the scores of students with a 50% likelihood of achieving a B average in a freshman-level course (Allen & Sconing, 2005). The NWEA MAP assessments were used in order to connect third through 11th grade student growth with these ACT targets. [ii]
4. ACT-Select Universities. The NWEA study also extended the predicted scores beyond a single set of college readiness benchmarks to a variety of institutions to which students might apply. This target represents the backward mapping of the RIT score based on the average ACT score reported by students who apply to the nation's top public universities.

Growth Rates

The Map Growth calculator contains three different rates of MAP growth, all of which are derived from the national growth norms published by NWEA (Northwest Evaluation Association, 2012). Whichever growth rate is selected, the calculator sets the next three year's projected RIT values that this rate would achieve.

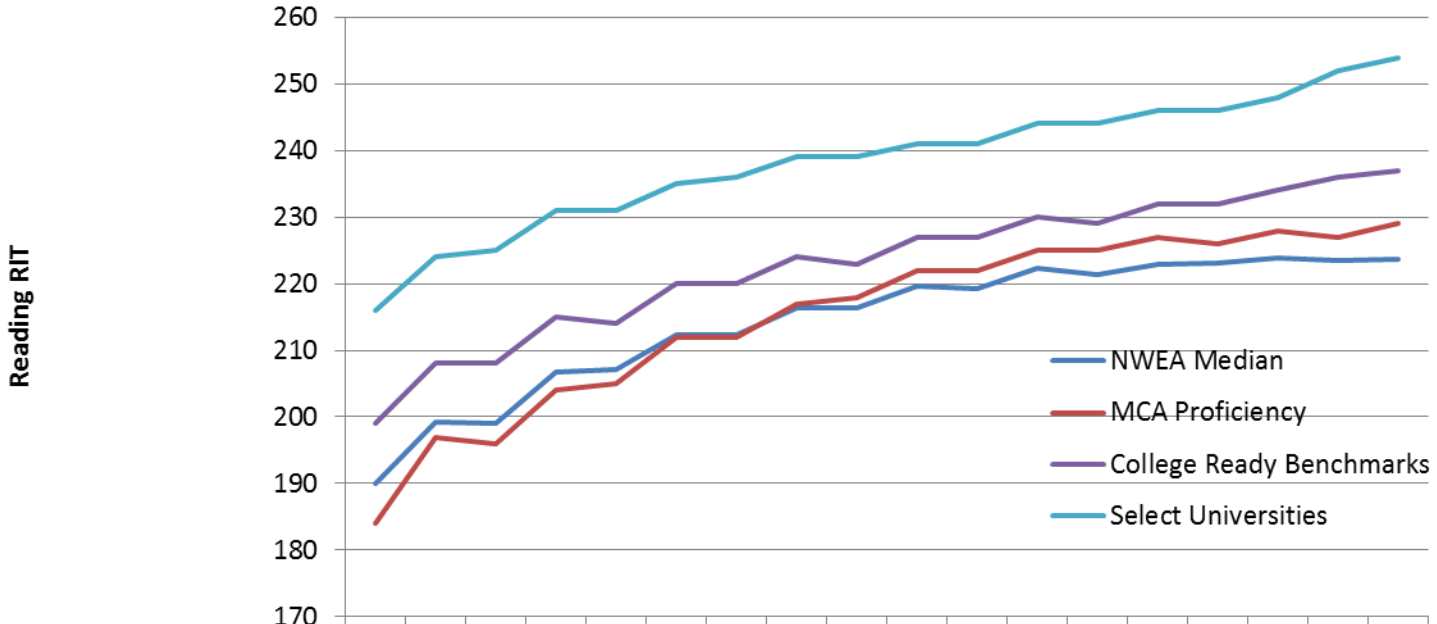
1. NWEA Average Growth. The growth displayed represents the projected increase in RIT score for the selected subject, grade and starting score over the next three testing seasons. [iii]

2. Ambitious Growth. The standard deviation of the mean growth was used to set a target that was approximately one-third of a standard deviation greater than average. Based on a normal curve equivalence, this rate could be described as growth at approximately the 63 percentile (Growth at the 50th percentile would be considered average).
3. Aggressive Growth. The standard deviation of the mean growth was also used to set a target that was approximately one-half of a standard deviation greater than average. Based again on the normal curve equivalence, this rate could be described as growth at approximately the 70th percentile.

Caveats and Cautions

"It's tough to make predictions, especially about the future." - *Yogi Berra*.
This quote should apply to an even greater extent when making predictions about young students across their entire academic trajectory. Factors like using predictors with less than perfect correlations; psychometric techniques of interpolation, extrapolation and smoothing; and estimating growth rates based on assumptions about normal growth all should make users of the MAP Growth Calculator cautious when it comes to making judgments about a student's future success. Clearly, there are many students who have been very successful in college with ACT scores that are less than the published benchmarks, and also many failures with scores that were above these benchmarks. The calculator and the resulting trajectories should only be considered guidelines of likely outcomes in MAP test scores given assumptions about growth and the known relationships to other assessments like the MCA and ACT. It should never be considered as a gatekeeping tool, nor should a student's plans be discouraged because he or she does not currently meet the selected target line.

Figure 1a gives the MAP Reading RIT scores for the four targets currently set in the MAP Growth Calculator. Figure 1b shows the corresponding percentile rank scores for these RIT scores for grades three and up. Figures 2a and 2b show the same information for the Math.



	3-Fall	3-Sp	4-Fall	4-Sp	5-Fall	5-Sp	6-Fall	6-Sp	7-Fall	7-Sp	8-Fall	8-Sp	9-Fall	9-Sp	10-Fall	10-Sp	11-Fall	11-Sp
NWEA Median	190	199	199	207	207	212	212	216	216	220	219	222	221	223	223	224	223	224
MCA Proficiency	184	197	196	204	205	212	212	217	218	222	222	225	225	227	226	228	227	229
College Ready Benchmarks	199	208	208	215	214	220	220	224	223	227	227	230	229	232	232	234	236	237
Select Universities	216	224	225	231	231	235	236	239	239	241	241	244	244	246	246	248	252	254

Fig. 1a Predicted Benchmarks in Reading as MAP RIT Scores by Grade and Season

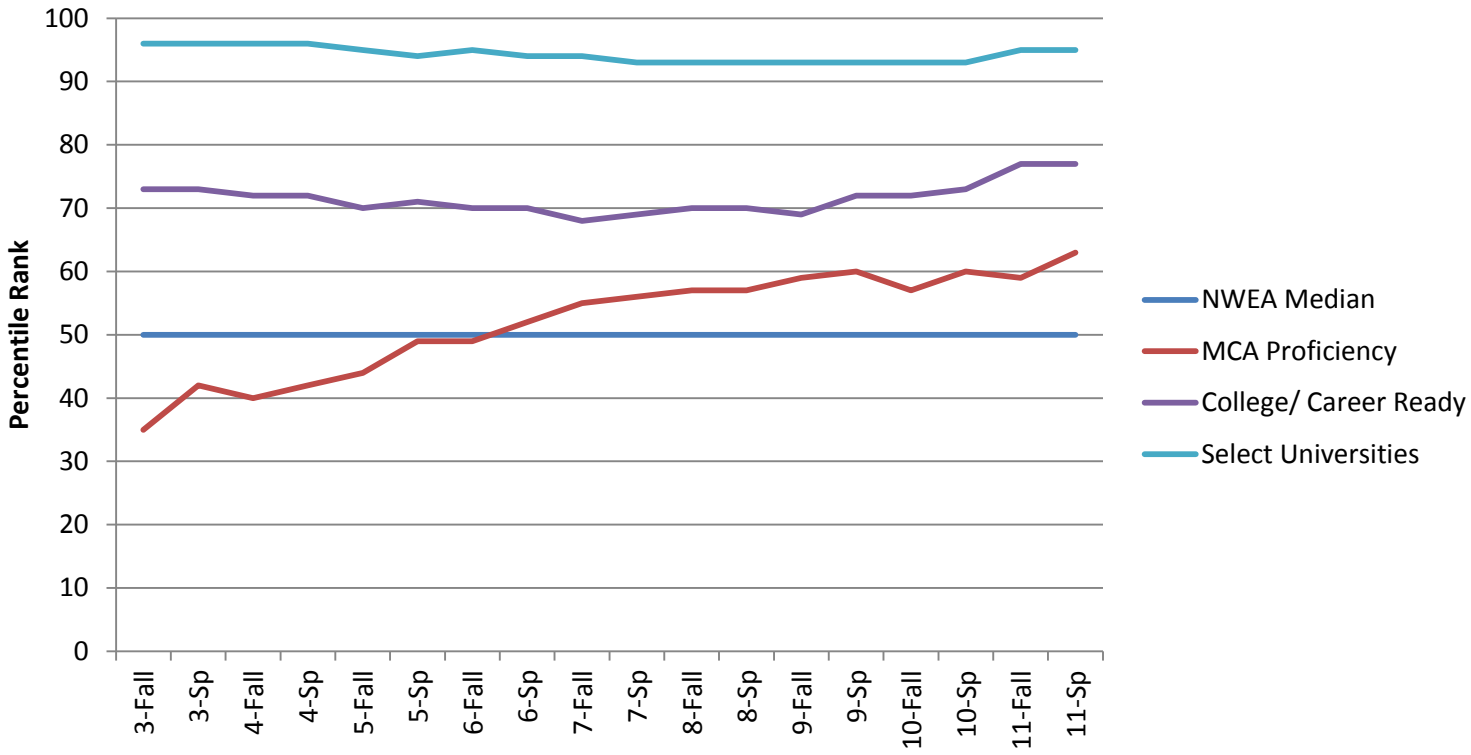
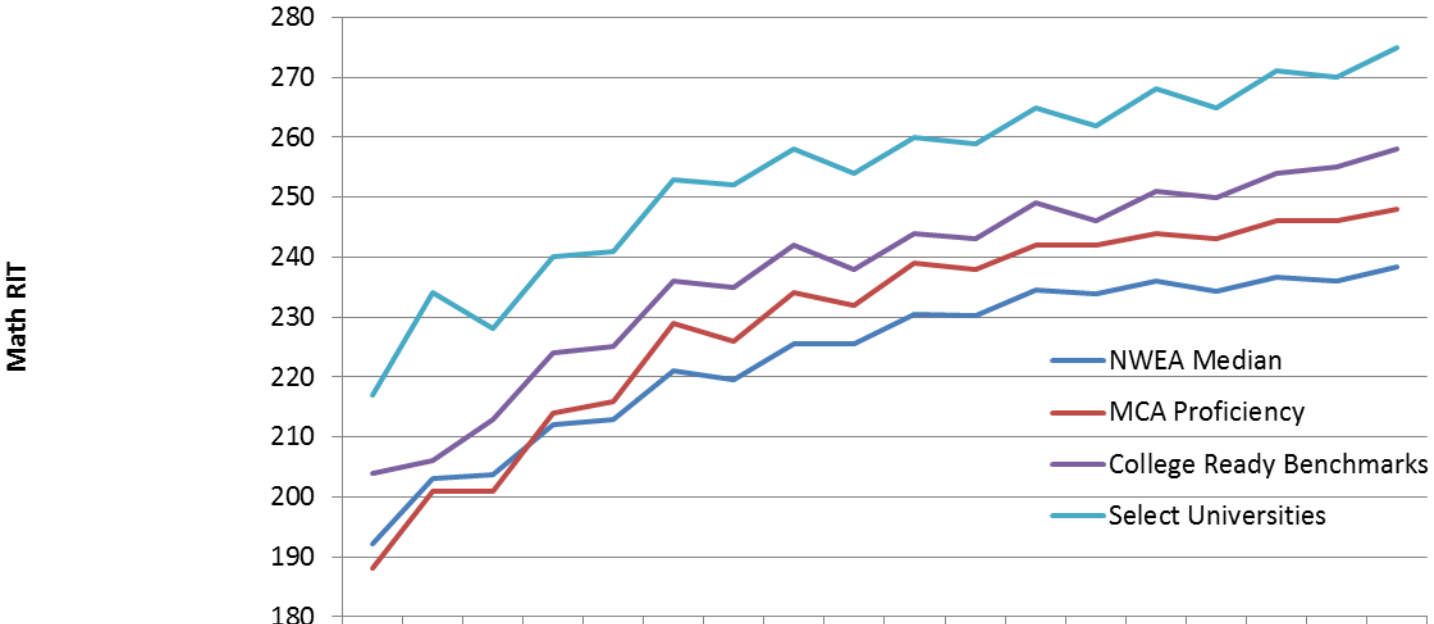


Fig. 1b Predicted Benchmarks in Reading as MAP Percentile Ranks by Grade and Season



	3-Fall	3-Sp	4-Fall	4-Sp	5-Fall	5-Sp	6-Fall	6-Sp	7-Fall	7-Sp	8-Fall	8-Sp	9-Fall	9-Sp	10-Fall	10-Sp	11-Fall	11-Sp
NWEA Median	192	203	204	212	213	221	220	226	226	231	230	235	234	236	234	237	236	238
MCA Proficiency	188	201	201	214	216	229	226	234	232	239	238	242	242	244	243	246	246	248
College Ready Benchmarks	204	206	213	224	225	236	235	242	238	244	243	249	246	251	250	254	255	258
Select Universities	217	234	228	240	241	253	252	258	254	260	259	265	262	268	265	271	270	275

Fig. 2a Predicted Benchmarks in Math as MAP RIT Scores by Grade and Season

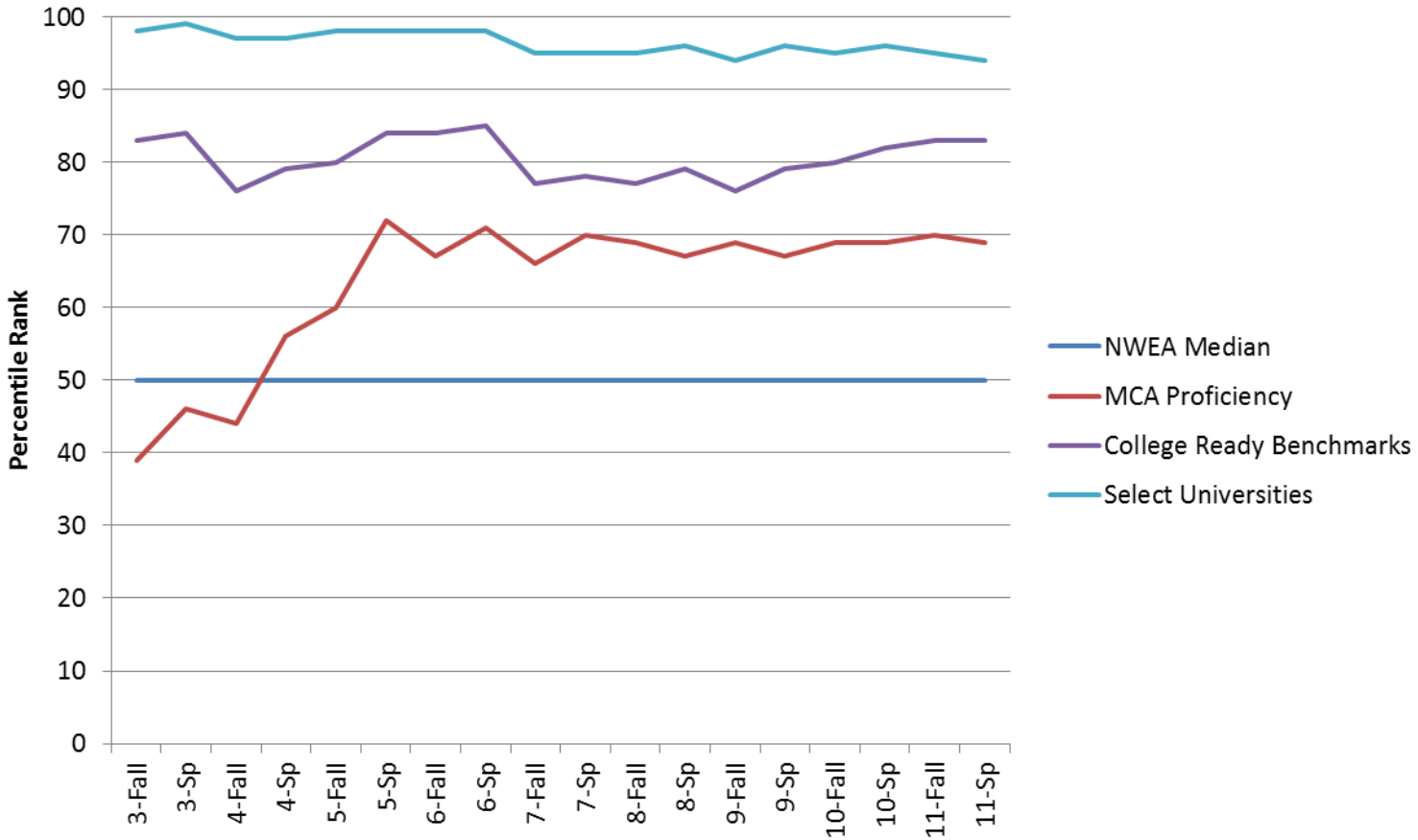


Fig. 2b Predicted Benchmarks in Reading as MAP Percentile Ranks by Grade and Season

References

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[i] These TIES studies were based on data from grades 3 through 8. The MAP Growth calculator extends these predictions through grade 11 by targeting a similar percentile rank score for the subsequent grades.

[ii] Since MAP data were only available beginning in grade 3, the targets for grades 1 and 2 were set using the same similar percentile rank score method described in note i.

[iii] Because of the way growth and status norms are calculated, a student who begins at the NWEA median RIT for a grade and grows exactly at the average growth rate over the next three years may not actually follow the NWEA target line, although the values should be quite close.